AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Orchid cultivation is highly dependent on environmental conditions as well as the health of the orchid plants. Therefore, managing orchid diseases is a major challenge for professional orchid growers and amateur orchid enthusiasts. Some common orchid diseases are caused by fungi, bacteria, or environmental pathogens. These diseases can be highly contagious and can have a significant impact on the growth patterns of orchid plants. In most cases, the identification of orchid diseases is based on manual observation by experienced orchid growers. However, this process is time-consuming and challenging for beginners. Although many existing approaches are based on the identification of orchid plant diseases using image processing or environmental monitoring techniques, most of the existing approaches are based solely on the identification of orchid plant diseases. The severity of the disease is not considered when determining treatment options for orchid plants, especially for Vanda orchid plants.

The research proposes AI-based Orchid disease detection, Severity assessment and Treatment recommendation system that integrates computer vision and environmental monitoring for orchid disease detection, severity assessment, and treatment recommendation. The system can be designed to analyze images of orchids to automatically detect any disease symptoms on the leaves, flowers, or roots of orchids. The system can also be designed to assess the severity of the disease. Environmental sensors can be used to collect data on temperature, humidity, light intensity, and moisture conditions to improve the accuracy of the system. The system can be designed to recommend appropriate treatments based on the severity of the disease detected to assist the growers in taking corrective actions.

The proposed solution uses deep learning for image analysis, machine learning for severity classification, and IoT for environmental sensing. The expected outcome is an intelligent decision support system that helps in early detection, correct treatment and reduction of losses. The social and commercial implications of the decision support system include a healthier orchid growth process, better quality flowers, reduced losses and a sustainable and profitable orchid cultivation process.

Keywords: Orchid disease detection, Vanda Orchids, Severity assessment, Treatment Recommendation, Computer Vision

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TABLE OF CONTENT

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 6
LIST OF FIGURES 7
LIST OF ABBREVIATIONS 8
LIST OF APPENDICES 9
1.INTRODUCTION 10
1.1Background and Context 10
1.2Literature Review 11
1.3 Research Gap 12
2.OBJECTIVE 14
2.1 Main Objective 14
2.2 Specific Objectives 14
3.METHODOLOGY 15
4.HIGH-LEVEL SYSTEM ARCHITECTURE 17
5USER REQUIREMENTS 20
6.COMMERCIALIZATION PLAN 22
7.BUDGET AND JUSTIFICATION 26
8.WORK BREAKDOWN STRUCTURE (WBS) 28
9.GANTT CHART 29
10.REFERENCES LIST 30
11.APPENDICES 31

LIST OF TABLES

Table 1: Research Gap Comparison.....12
Table 2:Cost-estimation for Orchid Disease Detection, Severity Assessment and Treatment Recommandation 24
Table 3: Pricing Strategy.....24
Table 4 : Budget and Justification.....26
Table 5: Gantt Chart.....29

LIST OF FIGURES

Figure 1: High Level System Diagram of AI-Powered Smart Orchid Care System Using Multi-Modal Machine Learning 17
Figure 2 : High Level System Architecture of Orchid Disease Detection, Severity Assessment and Treatment Recommendation 18
Figure 3 : Data Flow of the Orchid Disease Detection 18
Figure 4: Customer Persona 22
Figure 5: Revenue Model 23
Figure 6 : Work Break Down Structure 28

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
ESP32	Low-power microcontroller with Wi-Fi and Bluetooth
API	Application Programming Interface
DHT	Digital Humidity and Temperature Sensor
UI	User Interface
FR	Functional Requirement
NFR	Non-Functional Requirement
OpenCV	Open Source Computer Vision Library

LIST OF APPENDICES

Appendix 1 : Interview Questions 31
Appendix 2: Detailed System Architecture.....32
Appendix 3: UI Mockups.....32
Appendix 4:Risk Analysis.....33

1. INTRODUCTION

1.1 Background and Context

Orchid cultivation is an important aspect of the floriculture industry since orchids are highly valuable for their ornamental and commercial significance in both local and export markets. Among the various orchid species cultivated worldwide, the Vanda orchid is considered to be highly valued due to its bright colors, large flower size, and long pot life. However, orchids are highly sensitive to changes in the environment. Diseases such as leaf diseases, root diseases, stem diseases, and flower diseases are major threats to orchids. Studies on orchid intelligent cultivation and disease monitoring indicate that fungal infections, environmental stresses, and poor management conditions can greatly affect plant health and flower quality in an orchid cultivation systems [1] [2] [3]. If these diseases are not recognized in time, diseases can easily spread and affect the value and profit from the flowers.

This is a common problem for commercial orchid farms and nurseries, as well as for hobby growers. The impact for commercial growers is that diseases can cause severe financial losses due to poor quality flowers. The impact for hobby growers is the lack of expertise needed to properly diagnose and treat diseases. In most practical situations, orchid disease management still relies on observation and experience. Such an approach is often time-consuming and ineffective in the early stages of disease development [4] [3]. Research has shown that using an automated system to monitor orchid diseases can help reduce the burden of manual inspection in a greenhouse environment [4] [2] [3].

Most of the existing solutions are mainly used to detect simple diseases using plant images or environmental monitoring. For example, image-based methods with deep learning techniques have been proven to achieve extremely high accuracy in disease classification for orchids, while sensor-based methods can predict disease conditions by analyzing temperature and humidity patterns in greenhouses [2] [3] [4] [5]. However, it has been observed that many of these solution have some limitations. For example, these methods can only determine whether a plant has a disease, but cannot determine the severity of the disease, predict the future state of the disease, or suggest treatments for the disease based on the disease type and severity level [4] [5]. In addition, many sensor-based methods require IoT systems, hyperspectral imaging systems, or special equipment, which are expensive to maintain for ordinary growers [1] [5] [2] [3]. According to the reviewed literature, several studies on orchid diseases include disease prediction at the Dendrobium orchid, Phalaenopsis orchid, or greenhouse level, but there are few studies on intelligent disease assessment for Vanda orchid [4] [5] [2] [3].

This component aligns well with the Sustainable Development Goals, particularly SDG 12: Responsible Consumption and Production, as it contributes to the optimal use of plant protection resources by reducing losses due to early detection of diseases and providing treatment guidelines. It also aligns well with SDG 9: Industry, Innovation and Infrastructure, as it incorporates artificial intelligence, computer vision and sensor-based software technologies in orchid disease management. From an academic perspective, this study fits well with the SST - Software Systems and Technologies cluster, as it uses software intelligence, machine learning and monitoring technologies to solve a real-world problem.

1.2 Literature Review

Recent research on orchid health monitoring mainly focuses on disease detection, greenhouse disease prediction, pest identification, and smart farming support using artificial intelligence, computer vision, hyperspectral technology, and IoT technology [6]. The main research on health monitoring in orchids includes image-based disease classification. For example, the research paper "Chatbot-Powered Orchid Leaf Disease Classification and Management for Improved Farming" utilized the MobileNetV2 architecture to identify the presence of yellow leaf spot disease on the leaves of orchids. Furthermore, the identification is linked to a chatbot-based management platform [4]. In this case, deep learning is used to identify the presence of the disease, but the application is limited. Similarly, a lightweight hyperspectral deep learning model was developed to detect Fusarium wilt of Phalaenopsis from hyperspectral VNIR/SWIR imaging using EE-CSPNet, which enables high-accuracy detection of Fusarium wilt in Phalaenopsis orchids [5]. This research is important because it highlights the use of advanced imaging techniques for early disease detection, but it is based on the use of hyperspectral equipment, which is not only expensive but also not very practical for the average growers.

Another area of research is environment-based disease prediction. The paper “Designing a Spore Germination Sensor for Orchids” presented a model for sensor-based prediction of when the greenhouse environment is favorable for spore germination and disease prediction [2]. Another related paper, “Integrating a Spore Germination Sensor with Continuous Waveform Transforms for Orchid Disease Detection in Greenhouses,” further combined sensor data with time-frequency analysis and deep learning to detect rare orchid disease events [3]. These papers are important because they show that environmental factors such as humidity and temperature can be favorable for disease prediction. However, they mainly monitor outbreak conditions at the greenhouse level and do not analyze visible symptoms on the plant itself.

Research in orchid smart farming also includes integrated cultivation support systems. “OrchiZen Hybrid Integrated Smart Farming System for Orchid Plantations” combined IoT sensing, disease detection, smart irrigation, species recommendation, and fertilizer advice into one orchid-focused platform [1]. This is important because it shows the practical value of combining multiple technologies in orchid care. However, its disease component still mainly emphasizes detection, while severity analysis and treatment prioritization are not deeply developed. In addition, some orchid health studies focus on physiological stress and growth monitoring through advanced sensing. The study on multi-modal optical sensing fusion for orchid monitoring showed that combining machine vision, fluorescence, and hyperspectral methods can improve early stress detection and growth analysis [7]. Although very accurate, this type of approach is costly and suited mainly for large commercial greenhouse settings. Other AI-based orchid inspection systems also show that image-driven monitoring can support precision agriculture applications in orchid cultivation [8].

Overall, the literature shows clear progress in orchid disease detection, greenhouse monitoring, and orchid management support. However, most existing systems either focus only on disease presence detection, depend heavily on specialized hardware, or target species such as Phalaenopsis rather than Vanda orchids. There is still limited research on systems that combine

image-based disease identification, severity assessment, environmental context, and treatment recommendation in one intelligent workflow. This creates a clear research opportunity for the proposed component, which aims to deliver a software-driven, Vanda-specific disease intelligence system.

Limitations of Existing Systems:

- Most studies detect only whether a disease is present and do not assess disease severity levels.
- Existing systems rarely provide treatment recommendations based on disease type and severity.
- Many approaches depend on hyperspectral imaging, greenhouse devices, or specialized sensors, which increase cost and complexity.
- Several solutions focus on Phalaenopsis orchids or greenhouse outbreak prediction rather than Vanda-specific disease assessment.
- Commercial apps provide general advice, but they lack AI-driven severity analysis and predictive treatment support.

1.3 Research Gap

At the overall system level, existing orchid-care research has improved disease detection, greenhouse monitoring, and smart farming support, but most current solutions are still fragmented. Some studies focus only on identifying whether a disease is present from images, while others focus on greenhouse-level outbreak prediction using environmental signals. Very few systems combine image-based diagnosis, environmental context, severity assessment, and treatment recommendation within one orchid-specific workflow [1] - [5]. This creates a major gap in practical disease intelligence systems for growers.

For this component, the gap is more specific. Existing studies on orchid disease monitoring mainly focus on disease presence detection, early warning, or greenhouse infection risk, often for species such as Dendrobium and Phalaenopsis rather than Vanda orchids [2] - [5]. Although these systems report strong accuracy, they are still insufficient for real-world orchid care because they do not explain how severe the infection is, how it may progress, or what treatment should be applied based on the disease condition. In addition, many approaches depend on hyperspectral imaging, greenhouse-scale monitoring infrastructure, or specialized sensors, which reduce practicality for small-scale growers [1] [5] [2] [3]. There is still limited research on an intelligent, Vanda-specific system that can detect major diseases such as Phytophthora, Anthracnose, and Fusarium, assess severity, and recommend suitable treatments using both plant images and environmental sensor data.

Table 1: Research Gap Comparison

Existing Research / System	What it does	Research Gap	Features / Technologies Used
Chatbot-Powered Orchid Leaf Disease	Identifies whether an orchid leaf is healthy	Detects disease presence only; does	Image-based disease classification,

Classification and Management for Improved Farming [4]	or infected and reports results through a chatbot platform.	not assess severity or recommend treatment according to severity. Also focuses on yellow leaf spot disease.	MobileNetV2, transfer learning, chatbot platform, cloud-based reporting
CWT-IoT Device for Detecting Rare Events of Orchid Disease [9]	Detects greenhouse disease outbreak conditions from environmental data.	Predicts outbreak risk but does not diagnose visible plant disease, assess severity, or recommend treatment.	Environmental monitoring, Continuous Wavelet Transform, 3-D ConvLSTM, IoT-based data analysis
Detection of Phalaenopsis Fusarium Wilt Using EE-CSPNet [5]	Detects Fusarium wilt at an early stage in *Phalaenopsis* orchids using hyperspectral imagery.	Focuses on one species and one disease; does not grade severity or provide treatment guidance.	Hyperspectral imaging, EE-CSPNet, entropy-based attention, automated segmentation
OrchiZen Hybrid Integrated Smart Farming System [1]	Detects orchid disease and links it with broader smart farming features.	Includes disease detection, but severity-aware treatment recommendation is not deeply addressed.	YOLO-based disease detection, IoT sensors, smart farming integration
Design of a Spore Germination Sensor for Orchids [2]	Estimates greenhouse infection percentage based on temperature and humidity.	Focuses on greenhouse-level disease risk, not individual plant diagnosis or plant-level care recommendations.	Micro weather stations, germination modeling, threshold logic, environmental disease prediction
Identification of Pests on Black Orchid Plants Using Naïve Bayes Method [10]	Identifies pest attacks from leaf image texture.	Focuses on pests rather than orchid diseases, severity analysis, or treatment support.	Image preprocessing, GLCM texture features, Naïve Bayes classifier

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop an AI-assisted disease intelligence system for Vanda orchids that can detect Phytophthora, Anthracnose, and Fusarium, assess disease severity, and generate treatment recommendations using image analysis and environmental sensor data, achieving at least 90% accuracy in disease identification and severity classification.

2.2 Specific Objectives

- To identify and define the visual and environmental characteristics of Phytophthora, Anthracnose, and Fusarium in Vanda orchids using image samples and sensor readings.
- To design and implement an AI-based image analysis model that can automatically identify Phytophthora, Anthracnose, and Fusarium diseases in Vanda orchids from leaf, flower, and stem images under different greenhouse conditions.
- To develop a severity assessment mechanism that classifies disease infection levels in Vanda orchids into stages such as early, moderate, and severe, using both visual symptoms and environmental conditions.
- To integrate environmental sensor data with image-based analysis in order to improve disease diagnosis and severity prediction.
- To design and validate a treatment recommendation module that provides suitable care actions for Vanda orchids based on the identified disease type, severity level, and environmental context.

3. METHODOLOGY

This component is part of the overall AI-Powered Smart Orchid Care System, which combines image analysis, sensor readings, and machine learning to support disease diagnosis and decision-making for orchids. Within the overall system, this component specifically focuses on Orchid Disease Detection, Severity Assessment and Treatment Recommendation for Vanda orchids, with emphasis on Phytophthora, Anthracnose, and Fusarium disease types. The methodology is designed to move beyond simple disease detection by integrating visual and environmental information to support severity-aware and treatment-oriented disease management.

The research will follow a Design Science and Experimental Research approach. The Design Science aspect is used because the study aims to build and evaluate a software-based intelligent component for orchid disease management. The Experimental aspect is used to train, test, and validate machine learning models using image and sensor data under controlled and practical greenhouse conditions. Similar AI- and sensor-driven approaches have been used in previous orchid disease and smart farming studies, showing that image-based learning, environmental sensing, and integrated prediction models are effective for orchid monitoring [1] - [5].

From the technical perspective, the methodology begins with data collection and preprocessing. A labeled dataset of Vanda orchid images will be created to represent healthy plants and plants affected by Phytophthora, Anthracnose, and Fusarium, including multiple severity levels. In parallel, environmental sensor data such as temperature, humidity, light intensity, and moisture will be collected to capture the growth conditions surrounding each sample. Image preprocessing will include resizing, normalization, and enhancement of visible disease symptoms. Sensor data will be cleaned and synchronized with the image records before analysis. This multi-modal setup is supported by prior work showing the value of combining image-based features and environmental data for orchid monitoring and disease prediction [1] [2] [3].

The disease detection stage will use a computer vision model to identify the target orchid diseases from leaf, stem, and flower images. The severity assessment stage will classify infection levels into categories such as early, moderate, and severe. A treatment recommendation module will then map the detected disease type and severity level to suitable treatment actions. The overall system will therefore include three linked intelligence layers: disease identification, severity assessment, and treatment recommendation. Existing research has mainly focused on disease presence detection, while severity-aware treatment support remains limited [1] [4] [5].

The main development tools and platforms will include Python for model development, TensorFlow or PyTorch for machine learning, OpenCV for image preprocessing, and a suitable backend framework such as FastAPI or Flask for integrating predictions into the larger system. These tools are chosen because they are widely used in computer vision and machine learning research, support model training and deployment efficiently, and are compatible with software-based intelligent systems in the SST – Software Systems & Technologies cluster.

Data will be collected in a controlled and ethical manner. Orchid images and sensor readings will be obtained from greenhouse observations, public research datasets where available, and expert-supported labeling. Since the study does not involve human personal data, the main ethical concern is ensuring accurate data labeling and responsible use of agricultural observations. Expert validation will be important when assigning disease type and severity labels.

The system will be validated using standard performance measures such as accuracy, precision, recall, F1-score, and confusion matrices for disease classification. Severity assessment performance will also be evaluated against expert-labeled severity levels. The relevance of treatment recommendations will be assessed with expert review and practical suitability for orchid growers. Through this validation strategy, the study aims to demonstrate that the proposed component can provide accurate, practical, and early disease intelligence for Vanda orchids.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed AI-Powered Smart Orchid Care System is designed as an integrated platform with four main components: Orchid Disease Detection, Severity Assessment and Treatment Recommendation, Smart Watering and Nutrient Forecasting, Orchid Growth Stage Recognition and Bloom-Date Prediction, and Hybrid Pollination and Compatibility Analysis. These components work together to support the full care lifecycle of orchids by combining image analysis, sensor monitoring, and machine learning. In the overall system, the disease component plays a key role because plant health status directly affects watering decisions, growth analysis, and pollination suitability.

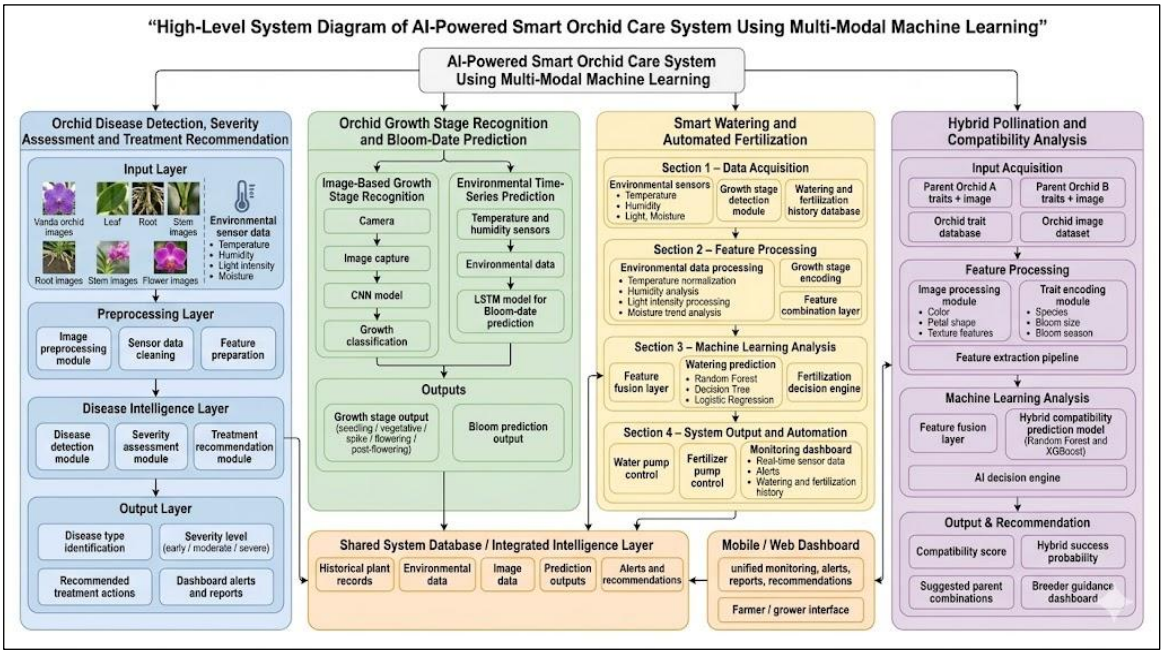


Figure 1: High Level System Diagram of AI-Powered Smart Orchid Care System Using Multi-Modal Machine Learning

This proposal focuses on the Orchid Disease Detection, Severity Assessment and Treatment Recommendation component for Vanda orchids. This component takes orchid images and environmental sensor readings as its main inputs. The image inputs include leaf, stem, and flower images, while sensor inputs include temperature, humidity, light intensity, and moisture conditions from the orchid-growing environment. These inputs are first sent to preprocessing modules, where images are cleaned and enhanced, and sensor values are normalized and organized.

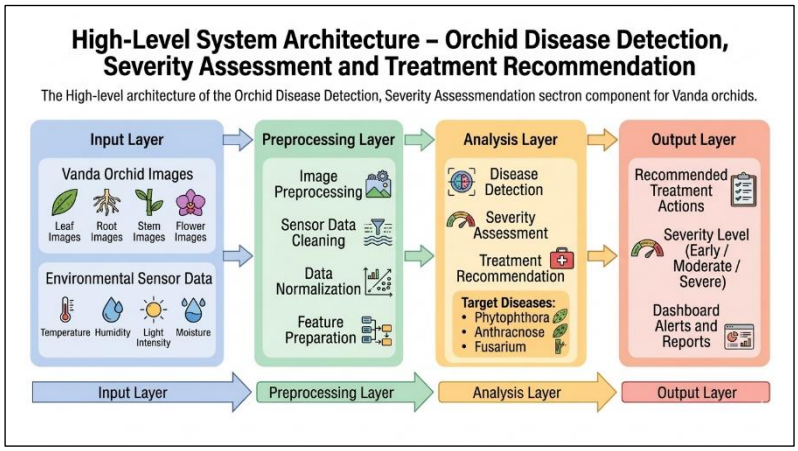


Figure 2 : High Level System Architecture of Orchid Disease Detection, Severity Assessment and Treatment Recommendation

After preprocessing, the component performs three main internal functions. First, the disease detection module identifies whether the orchid is affected by one of the target diseases: Phytophthora, Anthracnose, or Fusarium. Second, the severity assessment module determines the seriousness of the detected disease by classifying it into severity levels such as early, moderate, or severe. Third, the treatment recommendation module generates suitable actions based on the detected disease type, severity level, and environmental condition. These outputs are then delivered to the grower through the system dashboard.

This component also interacts with other parts of the overall system. Environmental data used here may also be shared with the watering and nutrient forecasting module. Growth-stage information from the bloom prediction component may provide supporting context for disease interpretation. In this way, the disease component works both as an independent module and as part of the larger orchid care platform.

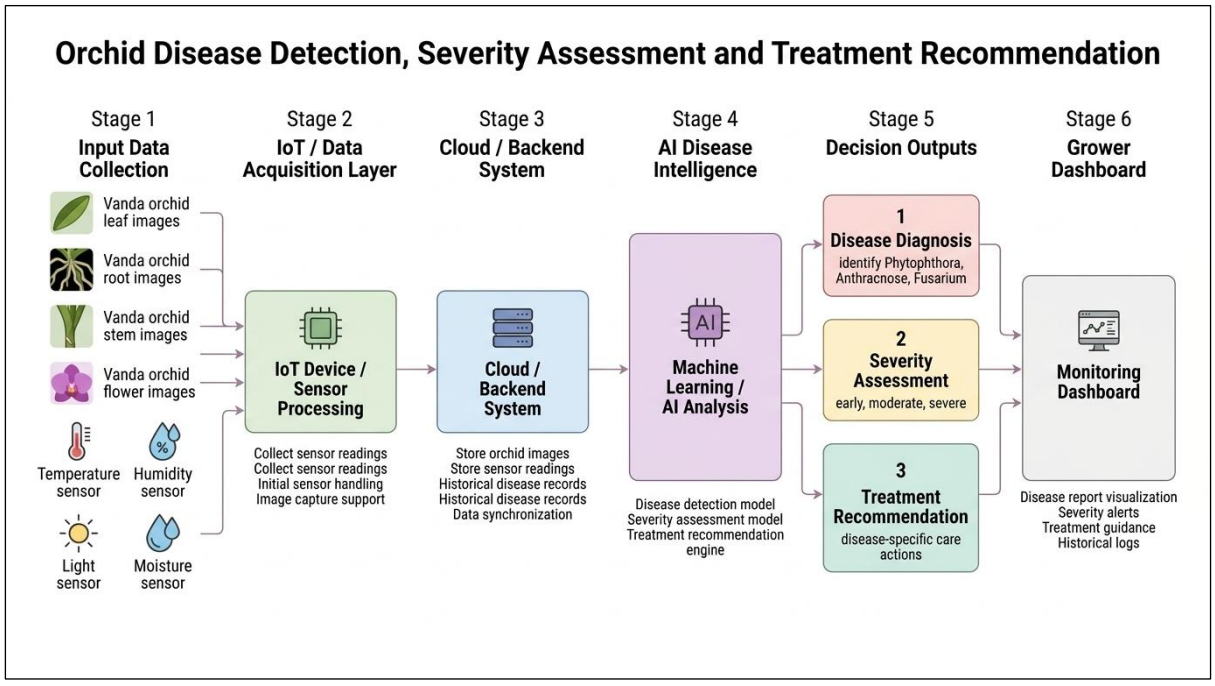


Figure 3 : Data Flow of the Orchid Disease Detection

The data flow begins with collecting Vanda orchid images and environmental sensor readings. These inputs are first preprocessed to improve image quality and organize sensor values. The processed data is then analyzed by the disease detection module to identify Phytophthora, Anthracnose, or Fusarium. Next, the severity assessment module determines whether the infection is early, moderate, or severe. Finally, the treatment recommendation module generates suitable actions, and the results are displayed through the grower dashboard.

The architecture is scalable because new orchid diseases, additional orchid species, or updated treatment rules can be added in the future without redesigning the full platform. Security is supported through controlled sensor access, safe storage of orchid images and records, and secure communication between the processing modules and user dashboard.

5. USER REQUIREMENTS

Stakeholder Identification:

Several stakeholders interact with the proposed Orchid Disease Detection, Severity Assessment and Treatment Recommendation component.

- Orchid Growers
  These are the primary users of the system. They use the system to detect diseases in Vanda orchids, understand the severity level, and receive suitable treatment recommendations to reduce plant loss and improve flower quality.
- Small-Scale Nursery Owners
  Nursery owners can use the system to monitor orchid health more efficiently without depending fully on manual inspection. This helps them take early action against diseases and maintain healthier plants.
- Home Orchid Enthusiasts
  Individuals who grow orchids at home can use the system to identify diseases and get proper treatment guidance without needing advanced expert knowledge in orchid care.
- System Administrators
  System administrators maintain the software platform, manage stored image and sensor data, monitor system performance, and ensure that the component operates securely and reliably.

Functional requirements:

FR1: The system shall allow users to upload or capture images of Vanda orchid leaves, roots, stems, and flowers.

FR2: The system shall collect environmental sensor readings such as temperature, humidity, light intensity, and moisture.

FR3: The system shall identify whether the orchid is healthy or affected by Phytophthora, Anthracnose, or Fusarium.

FR4: The system shall classify the detected disease into severity levels such as early, moderate, and severe.

FR5: The system shall combine image analysis results with sensor data to improve disease diagnosis.

FR6: The system shall generate treatment recommendations based on the disease type and severity level.

FR7: The system shall display the disease result, severity level, and recommended treatment through a dashboard or user interface.

FR8: The system shall store image records, sensor data, and diagnosis results for future analysis and monitoring.

Non-functional requirements:

Non-functional requirements describe the expected quality of the system in terms of performance, accuracy, usability, reliability, security, and maintainability.

- NFR1. Performance: The system should provide disease detection and severity results within a reasonable response time after image upload and sensor input.
- NFR2. Accuracy: The system should achieve high accuracy in identifying Phytophthora, Anthracnose, and Fusarium, as well as in classifying severity levels.
- NFR3. Usability: The user interface should be simple and understandable for both commercial growers and home orchid enthusiasts.
- NFR4. Reliability: The system should produce stable and consistent results when the same type of orchid data is provided under similar conditions.
- NFR5. Scalability: The architecture should allow future extension to support more orchid diseases, more orchid species, and additional treatment rules.
- NFR6. Security: Image records, sensor readings, and stored diagnosis results should be protected from unauthorized access.
- NFR7. Maintainability: The system should be easy to update when new disease data, severity rules, or treatment knowledge becomes available.

6. COMMERCIALIZATION PLAN

1) Target market and customer persona

The proposed component is mainly designed for the orchid cultivation industry, especially growers and businesses involved in Vanda orchid production. Since Vanda orchids are high-value ornamental plants, disease outbreaks can directly reduce flower quality, plant survival, and commercial income. This makes early disease detection and proper treatment support highly valuable for growers.

The main target users are:

- **Orchid Nurseries and Commercial Orchid Farms**
  These are the primary customers of the system. Large and medium-scale orchid farms can use the component to identify diseases early, understand severity levels, and apply correct treatments before serious losses occur.
- **Small-Scale Orchid Growers**
  Smaller growers often depend on manual observation and personal experience to identify orchid diseases. This component can help them make faster and more accurate decisions without requiring advanced expert knowledge.
- **Home Orchid Growers and Hobbyists**
  Individuals who grow Vanda orchids at home can use the system to detect diseases and receive treatment guidance, reducing plant loss and improving plant care.
- **Agricultural Experts and Floriculture Advisory Services**
  Experts and advisory services may use the system to support growers, validate results, and improve disease management practices in orchid farming.

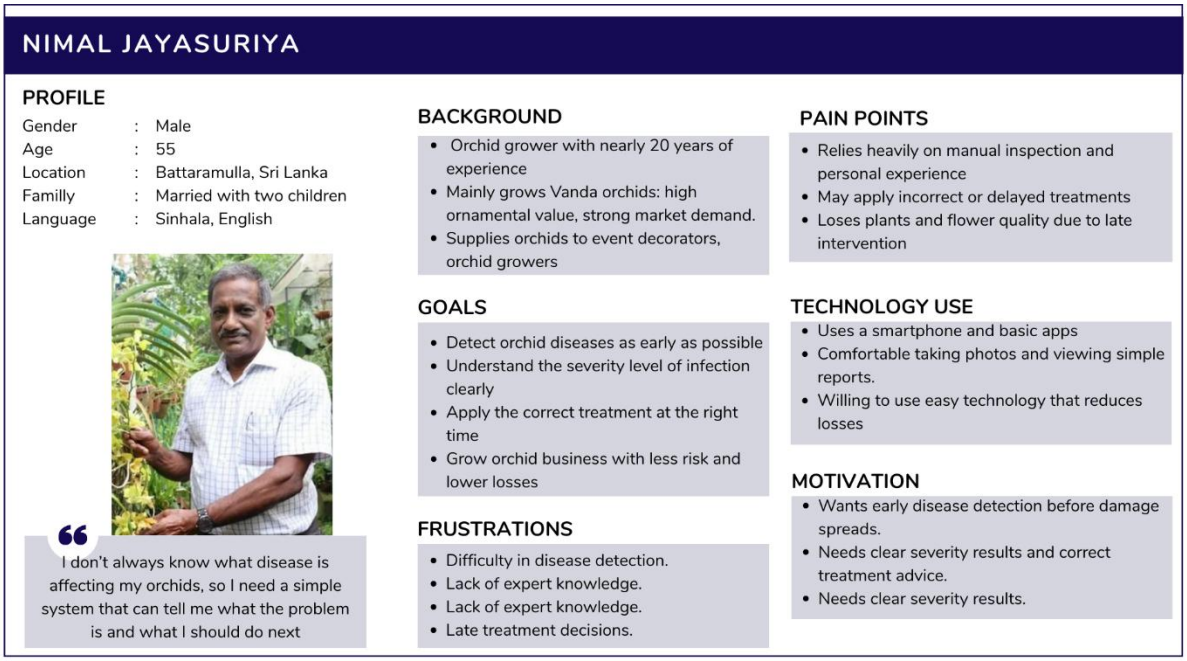


Figure 4: Customer Persona

2) Value proposition

The proposed component provides several important benefits to orchid growers:

- Early detection of Phytophthora, Anthracnose, and Fusarium in Vanda orchids
- Disease severity assessment, not just disease presence detection
- Treatment recommendations based on disease type and severity
- Improved flower quality and reduced disease-related plant loss
- Better decision-making using both image analysis and environmental sensor data
- Reduced dependence on expert manual inspection

Compared with traditional orchid care methods and basic plant disease apps, this component provides intelligent, severity-aware, and treatment-oriented disease management tailored specifically to Vanda orchids.

3) Revenue model

The component can generate revenue through a software-assisted smart agriculture model.

- Subscription-Based Software Service -
  Growers can pay a monthly subscription to access disease detection, severity assessment, and treatment recommendation features.
- Premium Diagnosis Service -
  Advanced features such as detailed history tracking, treatment logs, and sensor-based monitoring can be offered as premium services.
- Hardware-Assisted Package Sales -
  The system can also be offered together with a camera and environmental sensor package for growers who want a complete solution.
- Training and Installation Support -
  Revenue can also come from one-time installation, onboarding, and training services for farms and nurseries.

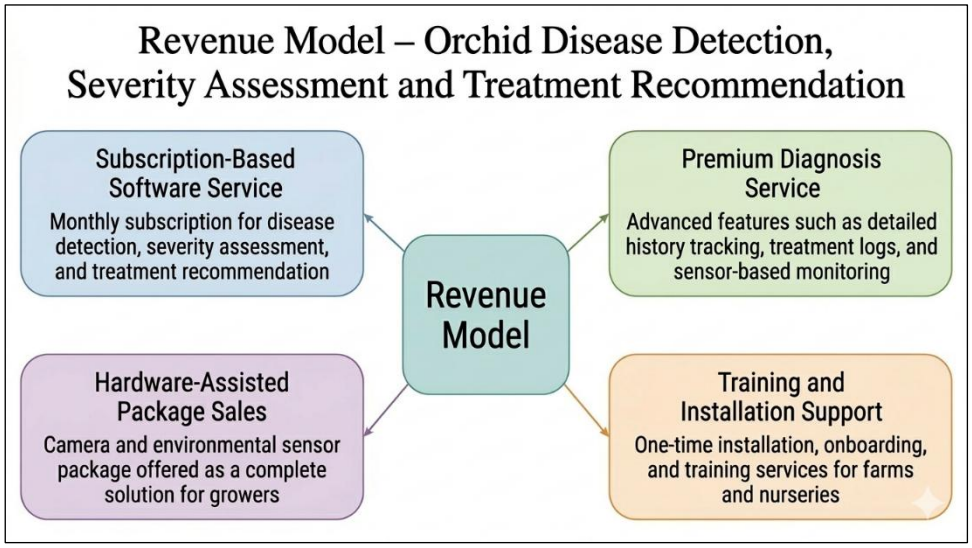


Figure 5: Revenue Model

4) Cost estimation

Table 2:Cost-estimation for Orchid Disease Detection, Severity Assessment and Treatment Recommandation

Item	Amount (LKR)	Description
Raspberry Pi Camera Module	4500 - 6000	Used to capture orchid disease images
DHT22 Temperature & Humidity Sensor	900 - 1200	Measures environmental conditions
Light / Moisture Sensors	1,000 - 2,000	Supports environmental monitoring
ESP32 Microcontroller	2000 - 3000	Collects sensor data and controls the system
Raspberry Pi Zero 2W	3500 - 4500	Used for image processing
Power Supply and Cables	2500 - 3500	Required hardware connections
Total Hardware Cost per Unit	14400 - 20200	Estimated cost for one system
Software Tools	0	Python, OpenCV, TensorFlow/PyTorch are open source
Cloud Services	0	Firebase free tier for data storage
Estimated Total Cost	**14400 - 20200**	**Approximate system cost**

5) Pricing strategy

- The proposed orchid disease detection component can generate revenue through a subscription-based service with optional sensor-supported features.
- Users can access the system through a monthly plan depending on the level of disease intelligence and support they need. A tiered pricing model is suitable so that both small-scale growers and larger orchid farms can afford the system.

Table 3: Pricing Strategy

Plan	Monthly Price (LKR)	Features
Basic Plan	500 – 1,000	Disease detection only
Standard Plan	1,000 – 2,000	Disease detection + severity assessment
Premium Plan	2,500 – 4,000	Disease detection + severity assessment + treatment recommendation + sensor integration

This approach allows users to choose a plan according to farm size and required features.

6) Competitive advantage

The proposed component offers several advantages compared with existing orchid disease management solutions.

- Uses AI to detect diseases in Vanda orchids more accurately than manual inspection.
- Assesses disease severity, not only disease presence.
- Recommends treatments based on disease type and severity level.

- Combines image analysis and environmental sensor data for better diagnosis.
- Provides a practical and orchid-specific solution for growers.

Most existing orchid disease systems mainly focus on basic disease detection or greenhouse-level monitoring. Many do not assess severity, do not provide treatment guidance, and are often designed for other orchid species such as Phalaenopsis rather than Vanda. The proposed component introduces a more complete disease intelligence approach by combining detection, severity assessment, and treatment recommendation in one system, making it more useful for practical orchid care.

7) Intellectual property considerations

- The main intellectual property of this component includes:
    - Trained machine learning models for orchid disease detection and severity assessment
    - Treatment recommendation logic based on disease type and severity
    - Software code for integrating image analysis, sensor readings, and decision support
- The software code can be protected through copyright, while the trained models, disease intelligence workflow, and treatment mapping logic can be protected as trade secrets or further explored for future patenting if the integrated workflow is considered sufficiently novel.

7. BUDGET AND JUSTIFICATION

The following table shows the estimated budget required to develop the AI Powered Smart Orchid Care System Using Multi-Modal ML.

Table 4 : Budget and Justification

Hardware Costs (per unit)			
Cost Category	**Estimated Amount (LKR)**	**Notes**	**Justification**
Raspberry Pi Camera Module	4,500–6,000	Captures plant images	Captures daily images of Vanda orchids to track growth stages (seedling, vegetative, spike, flower, post-bloom).
DHT22 Temperature/Humidity Sensor	900–1,200	Measures environment	Measures greenhouse environment. Temperature and humidity are key factors affecting flowering time and must be included in the prediction model.
ESP32 Microcontroller	2,000–3,000	Processes sensor data	Reads sensor data and sends it to the cloud via WiFi. Low cost and low power consumption.
Raspberry Pi Zero 2W	3,500–4,500	Runs image processing	Runs the image processing at the edge. It captures and analyzes images before sending results to the cloud.
Power supply, cables, SD card	2,500–3,500	Basic components	Provides stable power to the Raspberry Pi and sensors. Stores the operating system and temporary image data. Connects sensors to the board.
3D printed enclosure	1,000–1,500	Protects electronics	Protects electronics from greenhouse humidity, dust, and heat.
Total Hardware per Unit	14,400–19,700		
Software Costs			
Cost Category	Estimated Amount (LKR)	Notes	Justification
All software (Python, TensorFlow, etc.)	0	Open source	Free, open-source
Total Software	0		
Cloud Services (monthly)			
Cost Category	Estimated Amount (LKR)	Notes	Justification
Firebase free tier	0	Up to 1 GB storage	Provides database and storage for free up to 1GB, which is enough for a pilot project with one unit.
Total Cloud (Year 1)	0		
Human Effort Costs			
Cost Category	Estimated Amount (LKR)	Notes	Justification

Travel to greenhouses (fuel, bus fare)	5,000–8,000	10–15 visits	
Mobile data/Wi-Fi charges	5,000–8,000	For cloud testing and updates	
Refreshments during fieldwork	4,000–6,000	Snacks/meals during long days	
Total Human Effort [9]	14,000–22,000		

8. WORK BREAKDOWN STRUCTURE (WBS)

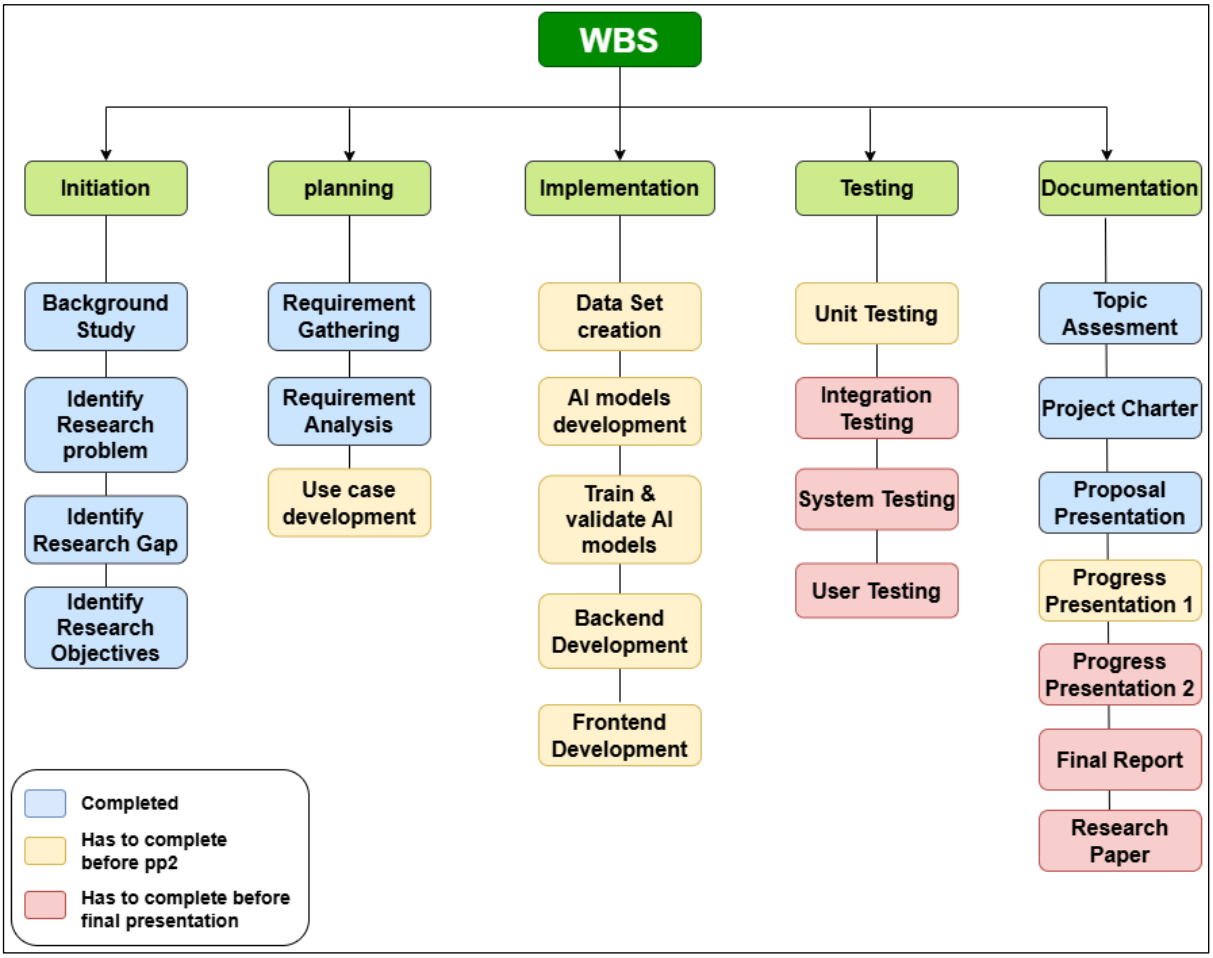


Figure 6 : Work Break Down Structure

9. GANTT CHART

Table 5: Gantt Chart

	Semester 1						Semester 2					
Activity	1	2	3	4	5	6	7	8	9	10	11	12
Preparation	■											
Identify the Problem	■											
Requirement gathering	■											
Literature review		■										
Analyze requirements		■										
Online project registration		■										
Topic assessment from submission		■	■									
Identify main objectives		■	■									
Project proposal submission			■	■	■							
Data Collection					■							
Model Training					■	■	■	■	■	■	■	
Software Development							■	■	■	■	■	
Progress presentation 1							■	■	■			
Integration									■			
Testing									■	■	■	
Progress presentation 2										■		
Research paper submission										■	■	
Progress presentation 2										■	■	
Finalizing												■

10. REFERENCES LIST

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11.APPENDICES

- Interview Questions

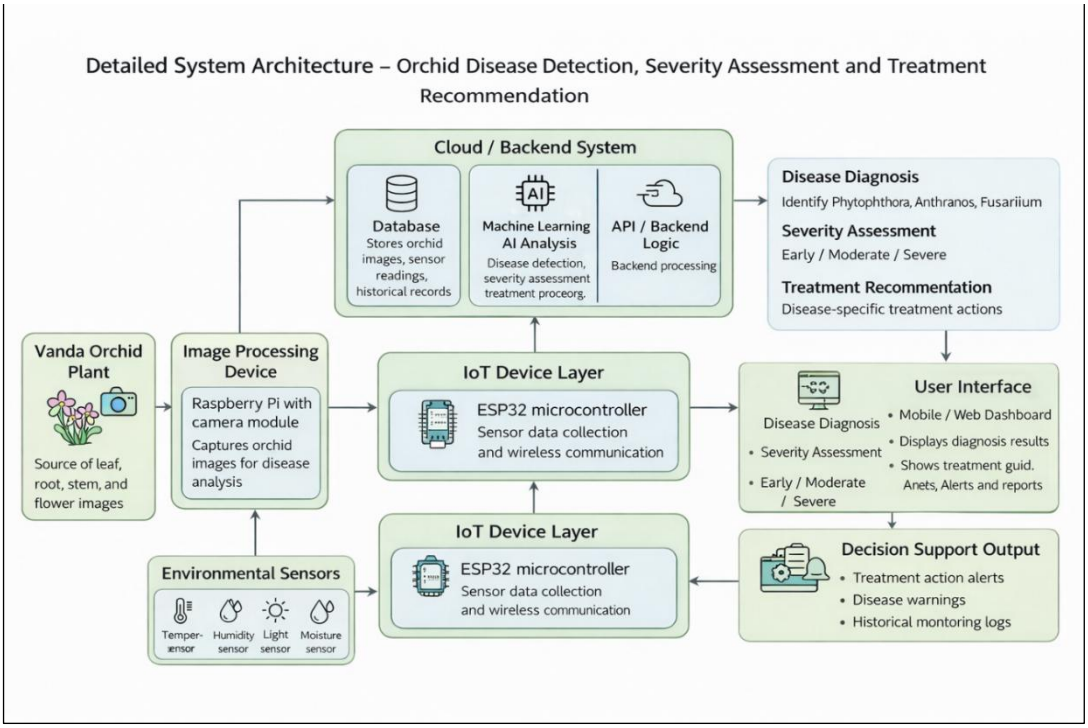
Appendix 1 : Interview Questions

Topic: Understanding current practices of orchid disease monitoring in Vanda cultivation

1. How many Vanda orchids do you currently grow?
2. What are the most common diseases you have observed in your orchids?
3. How do you currently identify whether orchid plants are diseased?
4. Do you usually check the leaves, roots, stems, or flowers when looking for disease symptoms?
5. How do you decide how serious a disease is once you notice it?
6. What actions or treatments do you normally use when a disease is detected?
7. Have you experienced problems with late disease detection or applying the wrong treatment?
8. Which disease causes the biggest loss in your orchid cultivation: Phytophthora, Anthracnose, Fusarium, or another disease?
9. Do environmental conditions such as humidity, temperature, light, or moisture affect disease spread in your greenhouse?
10. Would you find it useful if a system could automatically detect orchid diseases, assess severity, and recommend treatments?
11. What features would be most important to you in such a system?
12. How much would you be willing to pay monthly for a reliable orchid disease monitoring and recommendation system?

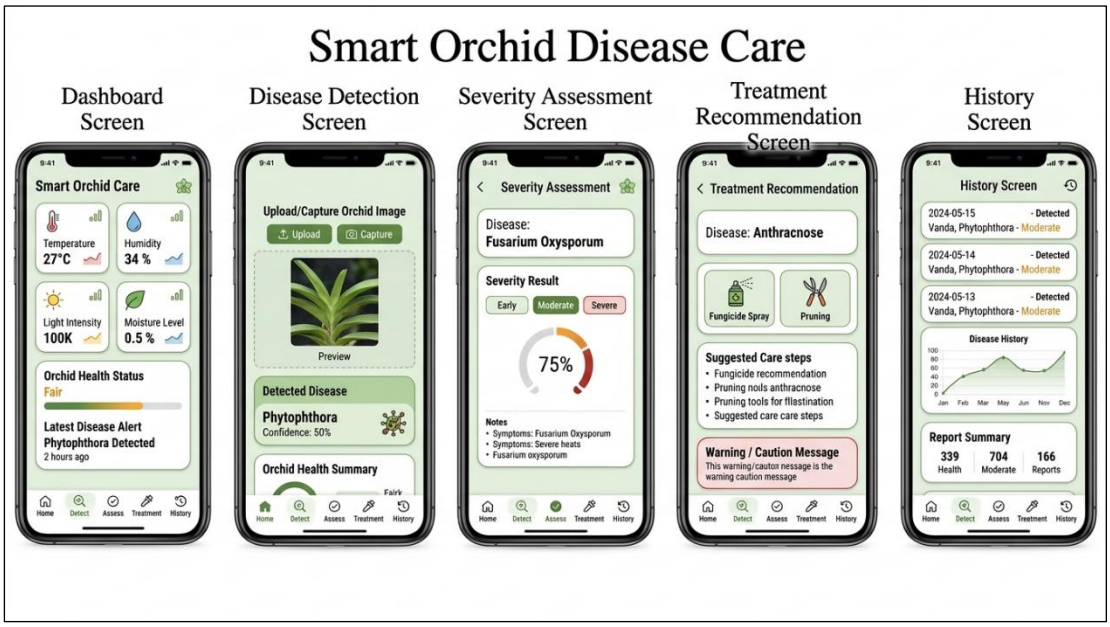
- Detailed System Architecture

Appendix 2: Detailed System Architecture



- UI Mockups

Appendix 3: UI Mockups



- Risk Analysis

Appendix 4:Risk Analysis

Risk Category	Identified Risk	Possible Impact	Mitigation Strategy
Data Risk	Poor quality or insufficient orchid image data	Low disease detection accuracy	Collect more labeled Vanda orchid images under different disease conditions and severity levels
	Incorrect or inconsistent expert labeling	Wrong disease or severity prediction	Validate labels with domain experts and cross-check samples before training
Hardware Risk	Sensor failure or inaccurate sensor readings	Incorrect environmental input to the model	Regular sensor calibration and periodic hardware checking
Model Risk	Disease detection model misclassification	Wrong disease diagnosis	Train with balanced datasets and evaluate using accuracy, precision, recall, and F1-score
Integration Risk	Failure in combining image and sensor data	Reduced system reliability	Test data synchronization and integration thoroughly before deployment
Usability Risk	Growers may misunderstand results	Wrong treatment decisions	Provide clear dashboard labels, alerts, and simple explanation of results
Security Risk	Unauthorized access to image records and diagnosis results	Data privacy and misuse issues	Use secure storage, controlled access, and authentication mechanisms